MIME Pilot whole Muscle EDL

Date: 2025-07-10
Tags: MIME, Pilot skeletal muscle active injury EDL
Created by: Sven Weidner

---

author: M.Millard
date: 21/08/2025

General Notes:

1. Experiment done on July 10, 2025
2. People involved
    1. Andre Tomalka performed the dissection
    2. Sven Weidner coded the pre-screening protocols and ran the experiment (hand made)
    3. Matthew Millard coded the normalization and injury protocols and ran the experiment. These files were generated using
        1. https://github.com/mjhmilla/AuroraCommandFileToolkit
        2. project_mime branch
        3. main/main_generateProjectMimeProtocols610A.m
        4. Note: the output from this code is in the normalization_protocol.zip and injury_protocol.zip files
3. Equipment used: Aurora 1200A small rodent testing machine (https://aurorascientific.com/products/muscle-physiology/systems/1200a-isolated-muscle-system-rodents/)
4. File types:
    1. *.dpf: command files to run the Aurora 1200A. Readable as a plain text file
    2. *.ddf: output files from the Aurora 1200A. Readable as a plain text file. Note that these files contain calibration data, the protocol, and the measurements from the Aurora machine.
    3. Note: a Matlab code repository is currently being written to read the ddf files and segment labels files to make it easier to work with this data.

Experiment-Specific Notes:

- The EDL was collected before the SOL on the same day.
- This was the first time that we tested a whole muscle and unfortunately it degraded quickly. We did not give the muscle enough time between stimulations to recover. Due to our ignorance, we also failed to perform two tasks that are known to help the muscle recover: gently cycle the length of the muscle, and oxygenate the solution. As such, perhaps the early trials are useful, but the rest should be approached with caution.

Overview:
This experiment has two blocks:

1. normalization
    1. Output files from the Aurora 1200A system begin with "normalization_"
    2. The protocol is in normalization_protocol.zip. In this zip file please see:
        1. See protocol_20250110.csv for the trial order and over view
        2. See metaData_20250710.csv for a blank file containing the meta data

        3. See instructions_20250710.csv for experimental instructions to be used during the collection.
        4. Each trial has a corresponding file in the segmentLabels folder. In each of these files the blocks within each trial are given a name, a starting time, and an ending time
2. injury
    1. Output files begin with "injury_"
    2. The protocol is in injury_protocol.zip. The structure of the contents in this folder are the same as for normalization

The normalization trials are used to measure

1. Length at which the maximum isometric force (fmax) is reached (lref)
2. Extrapolated length at which overlap is lost (lzero)
3. Optimal fiber length (lceOpt)
4. Length at which a passive force of half fmax is developed (lpeHalf)
5. The maximum shortening velocity (vmax)
6. The impedance of the muscle at lceOpt in the passive and active state

The injury block trials come in three parts:

1. Characterize the properties of the muscle: trials 01-10
2. Acutely injure the muscle with an active lengthening trial: trial 11
3. Characterize the properties of the muscle: trials 11-21

This protocol is being developed so that we can get a picture of how the mechanical properties of the muscle are affected by a partial injury as a part of project Muscle Injury Mechanics and Experiments (MIME). Financial support is gratefully acknowledged from the Deutsche Forschungsgemeinschaft through project 540349998.

Attached files

stimulation_long.dpf (Protocol: Isometric contraction under isometric force)
sha256: 22e1c589d45fa0e508446db71cc345fa1c088c8e4227bcd7cc4e26f6697549df

stimulation_ramp.dpf (Protocol: Similar to FLR_twitch, but instead tetanus stimulations (0.2s) rather than a twitch)
sha256: 3d2e8334c2d2489987794e818ebb137b487ecbc7216788336200327e88ba0a82

stimulation.dpf (Untitled)
sha256: 4a4e5cece480f1f413cfed1c4fea169162649d5619f04c919aaf82edfb8b0a6a

sine_cycling.dpf (Protocol: passive cycling of the length (recovery phase cycling))
sha256: 502b514759f8c57ecfdf50595aca68f2ef4f169cbc682095d3d2e23bf3cf0570

passive_ramp.dpf (Protocol: to measure the force-length relation)
sha256: 597a563c3fbf0fa9fba76a9e1dfdd6c60d46065e8510618bf766e45b6acd9f43

FFR_tetanus.dpf (Protocol: ramp protocol used to measure the force-length relation) sha256: f09801d753e6bc4fadf062e3a8720bbbc0bcede916680afaaaf47a7d97dd2a7c

FLR_twitch.dpf (Protocol: ramp protocol used to measure the force-length relation but with twitches rather than tetanus)
sha256: ec12be9b4fbb68b3b0fb58632314c9a0d02a9b1824052c2ed5be56f4fb9e93fa

sine_cycling_2.ddf (Data: passive cycling of the length (recovery phase cycling))
sha256: 5c9232955e64902003857a06598879c1ec9fd88d298d0ad1c7eb0af6ccf9a160

FLR.ddf (Data: force-length-relation)

sine_cycling_1.ddf (Data: passive cycling of the length (recovery phase cycling))
sha256: 4a7e45eaf7fc3b22da9b1fa404c7ba10ac0e8db1f68ec60f3c6e83e457f8af99

FFR.ddf (Data: force-frequency-relation)
sha256: bdef51d424cd143f8ecdeb5451bd63738fc772e611f238de80dbf94c153017c4

normalization_protocol.zip
sha256: 6d5772d36976173e80d54545ed42d436f26b147f6dc831375da11cbf7c1436e4

injury_protocol.zip
sha256: ba95a282d54d132aa722529200c176e292c89cae6c130c06e2eaf1dbe23b8260

normalization_05_forceVelocity_00Lo_20250710_EDL_half.ddf
sha256: 4ae36a123fda641801f37ec1a6b6ca0ab33303cef225d809ab7770e902b0a254

normalization_05_forceVelocity_00Lo_20250710_EDL.ddf
sha256: cdc8bec7a662c11b2d04ad66f595f8c11f99e83e7582214d35a794614997a5fb

normalization_03_twitchForceLength_20250710.ddf
sha256: 6ffcf48cd4fb14496abb2977ce086d53293ea98b29bd5659c09cd9c02f834783

normalization_04_isometric_20250710.ddf
sha256: 2b329f58baf49e5afffe1a7d66487a0be8fe3e28fdb126db0cafb0b76fce897d

normalization_07_impedance_00Lo_20250710.ddf
sha256: 5be014afa03d3c45c6453a3910b65a3f50f034c5a9794a1457bec09a4bcb2428

normalization_01_isometric_20250710.ddf
sha256: 76b0c309a14734061caa25cc27c42603f8a564fe26778514710c7d54a3cfbdac

injury_11_1_injury_09Lo_20250710.ddf
sha256: e8ca8b2eea881ba5838114af8a4edf6eed16a38c91911c5eda93cf4cbaf03824

injury_10_activeLengthening_09Lo_20250710.ddf
sha256: 70f5fb26b86616a3d4042a22dffc8d2d7bd8997a4def2d407fdf6f69ac6eb650

injury_09_activeLengthening_09Lo_20250710.ddf sha256: 9b19c666762e27751827a4820c6ff02216a328c5ab76afe55528aae7a152dcf0

injury_06_isometric_14Lo_20250710.ddf
sha256: a66a6682110db0a5e9a32cc8ac87dcbf3c0e6da50e27dfee89b757ff2c57c809

injury_05_isometric_10Lo_20250710.ddf
sha256: 47b35554fc910a7392a095f3d0e8bf9cd7886f8e42415407a848b7b2a1df8912

injury_07_activeShortening_11Lo_20250710.ddf
sha256: 0bfe5597dd165940e47c2a95beca2e3cbc3ec6e3221f24187f3c31efe62dfd60

injury_08_activeShortening_11Lo_20250710.ddf
sha256: 141674bf4c2da1709404f5fade9ef6df4a3ae8d5c14d0fdf7a1c94d68ba5f7a7

injury_04_isometric_06Lo_20250710.ddf
sha256: 14a49811c42373576c38f1fc6597b93875531163628a5bd89a82c2d6925926f0

injury_03_passiveLengthening_06Lo_20250710.ddf
sha256: 395b5ef19493b46d08f53f89b42737920cfca1157388943d8019f3471876c411

injury_02_passiveLengthening_06Lo_20250710.ddf
sha256: a62907ccd99202d07f1f0bec2f6b422ed528441e19251cbe2948f0898a4123ee

injury_01_isometric_10Lo_20250710.ddf
sha256: 3a70cf90b5b448d3ace0f387b1832d0e12dea6bd50a8c696015416616511bd94

injury_12_isometric_10Lo_20250710.ddf
sha256: 6c433e8a21a0dc7dec21d4f0211de113f6839f1b6ff900940188f08c3d0ecd85

injury_11_injury_09Lo_20250710.ddf
sha256: 3bcf196364ac91a79d9517dba77b2a8bd06ab80913082b1e9ae0d67c5eb2e1be

injury_12_1_isometric_10Lo_20250710.ddf
sha256: d65682ecea0339e90baaedd676fe551de9104ea627df480093e1e0229b7b1604

injury_11_2_injury_09Lo_20250710.ddf
sha256: 14a102744fc8bdb52fe446005c1ccbe5a2a00e6187fbee3db0fa6c20a470fab0

Unique eLabID: 20250821-953d8c5d918b3a65d4846d699d0b321d74837f95
Link: http://elbustest.uni-stuttgart.de/experiments.php?mode=view&id=223